CHAPTER 5

Statistical Model Selection: An Alternative to Null Hypothesis Testing

Alan B. Franklin, Tanya M. Shenk, David R. Anderson, and Kenneth P. Burnham

Model selection is an alternative statistical approach to null hypothesis testing when we wish to evaluate the plausibility of competing hypotheses or models. The primary advantage of the model selection approach is that it allows simultaneous examination of multiple working hypotheses rather than just two (the null and the alternative), as in null hypothesis testing. Moreover, model selection eliminates the need for an uninformative null hypothesis of "no difference" (although it can still be included in this approach). Finally, the model selection approach allows multiple hypotheses to be ranked and quantitatively evaluated in terms of their plausibility in explaining an empirical set of data and, as well, estimates of precision can incorporate the uncertainty about which model best approximates the data.

This chapter presents an overview of model selection based on Kullback-Leibler information (Kullback and Leibler 1951) and likelihood-based inference to evaluate the evidence for alternative scientific hypotheses. The information presented here is not new. For details of the philosophies and

mechanisms of model selection see Linkhart and Zucchini (1986), Miller (1990), and Burnham and Anderson (1998). What we present here is a primer on the model selection approach, an outline of the works just cited, and an example of how to apply the approach in analyzing empirical data.

The Development of Null Hypothesis Testing

Statistics is a language of science (Zimmerman 1995) that helps us in understanding the natural world. There are different branches of statistical language and diverse interpretations of these branches. For example, there are frequentist and Bayesian branches in statistics (Malakoff 1999; see also Chapter 3 in this volume), each with its own paradigms and philosophies. In general, frequentist statistics has dominated the field of applied statistics.

Null hypothesis testing has been important in the frequentist branch of statistics. Anyone who has taken a course in statistics has been exposed to null hypothesis testing. Indeed, this paradigm has dominated statistical thinking since its inception in the 1920s and 1930s (Royall 1997). Under the paradigm of null hypothesis testing, there are two main philosophies. The first philosophy, attributed to R. A. Fisher (1925), focuses on significance testing, where the degree to which evidence against a null hypothesis is measured in terms of a *p*-value. The second philosophy, developed under Neyman and Pearson (1933), focuses on the decision aspect of hypothesis testing—that is, whether to reject one of two hypotheses in favor of the other and to do so in terms of Type I and Type II error probabilities. The decision of which hypothesis to support is made according to whether a test statistic falls into a predefined critical region α (usually $\alpha = 0.05$). The result is a decision: "choose H_1" or "choose H_2" (Royall 1997: 63–64). Fisher's philosophy differs from Neyman-Pearson's in two fundamental ways. First, under Fisher's philosophy only the null hypothesis is considered and the alternative hypothesis (H_A) is simply something other than the null. In reality, only the null hypothesis is being tested because the null is the hypothesis of "no effect" (Royall 1997:63); thus one only tests the hypothesis of "no effect" against every other possibility. Second, Fisher's significance testing involves a *p*-value (Fisher 1958) that has been misused as a measure of evidence against the null hypothesis—for example, small *p*-values have been interpreted to mean the data provide evidence against the null hypothesis, whereas large *p*-values do not provide evidence against the null hypothesis. (See Graybill and Iyer 1994:32.) But *p*-values are an inappropriate metric for strength of evidence (see Royall 1997:8–13, 68–71). There is a key point here: even when the Neyman-Pearson and Fisher significance testing philosophies are combined (Schweder 1988), only two hypotheses can be examined simultaneously under the paradigm of null

hypothesis testing. Although multiple tests can be conducted (H_0 versus H_{A1}, H_0 versus H_{A2}, H_0 versus H_{A3}, and so forth), there is no way to rank hypotheses quantitatively when there are more than two *p*-values.

In addition to his contribution to significance testing in statistics, Fisher also introduced maximum-likelihood estimation in the 1920s. The idea of likelihood differs from probability in statistics in a fundamental way (Edwards 1992) with respect to how hypotheses relate to the empirical data. A hypothesis begins as a concept that can be expressed verbally—such as "the density of cottonwood seedlings decreases as distance from the river's edge increases." This conceptual idea can then be translated into a statistical model. For example, the preceding verbal hypothesis can be formalized as a regression model,

$$Y = \beta_0 - \beta_1 X$$

where Y is the density of cottonwood seedlings measured on plots at varying distances from the river's edge and X is the distance of each plot from the river's edge. The parameters of this model are β_0 (the intercept) and β_1 (the slope). Standard null hypothesis testing would test each parameter of the model separately: H_0: $\beta_0 = 0$ versus H_A: $\beta_0 \neq 0$ for the intercept and H_0: $\beta_1 = 0$ versus H_A: $\beta_1 < 0$ for the slope (Ott 1993:499–500). The interpretation of the *p*-values resulting from these tests is that they are the probability of observing a sample outcome more contradictory to H_0 than the observed sample result (Ott 1993:231) if H_0 was true. In null hypothesis testing, statistical probability models are based on $P(X|\theta, \text{model})$—the probability of obtaining data X given the parameter θ from the hypothesized statistical model (after Edwards 1992:9). Neither the idea of testing each parameter of the model separately nor the logical interpretation of resulting probabilities is intuitively appealing. Likelihood, by contrast, is expressed as $\mathcal{L}(\underline{\theta} \mid X, \text{model})$—the likelihood of the model parameters, $\underline{\theta}$, given the data X and the specific statistical model. The interpretation of likelihoods is more logical because the hypothesis is evaluated given the data rather than the data being evaluated given the hypothesis. This interpretation is carried through the model selection paradigm.

Kullback-Leibler information (Kullback and Leibler 1951) is the fundamental basis for the model selection paradigm. Conceptually this quantity, denoted as $I(f, g)$, is the information lost when model g is used to approximate full reality (f). Clearly we would like to use a model that loses as little information as possible as a basis for formal statistical inference. But we have to know reality (f) in order to compute Kullback-Leibler information in real-world situations. Obviously, we never know reality in natural resource (or any other scientific) application. A breakthrough came in 1973 when Hirotugo Akaike derived an estimator of Kullback-Leibler information (Akaike 1973).

His estimator was based on a bias correction of the log-likelihood function at its maximum and is called AIC for Akaike's Information Criterion. Thus likelihood theory is central to model selection (estimation of Kullback-Leibler information), parameter estimation (such as maximum-likelihood estimates), and hypothesis testing (such as likelihood-ratio tests).

The Idea of Multiple Working Hypotheses

Thomas C. Chamberlin was a geologist and president of the American Association for the Advancement of Science at the end of the nineteenth century (Hilborn and Mangel 1997). In a seminal paper (Chamberlin 1890) that was largely overlooked until the 1960s, he argued that scientists tend to have certain hypotheses that they subjectively favor and tend to promote in their work. In adopting a favorite hypothesis, scientists tend to filter their view of the world in favor of this hypothesis and, when faced with contradictory data, tend to reject the contradictory data rather than challenge their favorite hypothesis. This phenomenon simply points out that scientists are human: when faced with an explanation that seems logical and satisfactory, it is only human to want to believe in it. To avoid this subjective bias, Chamberlin suggested that creative scientists should consider a number of viable alternative explanations in the form of scientific hypotheses (hence the phrase "multiple working hypotheses") and should even consider alternatives that contradict their own beliefs if such candidates are feasible and realistic.

Platt (1964) argued that this idea of multiple hypotheses was both an explanation and a cure for many of the problems in science and indeed was at the heart of an approach to science he called *strong inference*. In strong inference, a number of alternative hypotheses are devised and then crucial experiments are set up such that many of these candidates can be excluded. Strong inference is not a requisite for doing science, but it is a requisite for rapid progress in science. In the study of natural resources and ecology, we are probably more guilty than most other branches of science in not following the tenets of Chamberlin and Platt (Romesburg 1991). At this point, a reasonable question arises: How do we incorporate the idea of multiple working hypotheses into ecological and natural resource sciences? This question was recently addressed by Hilborn and Mangel (1997) and by Burnham and Anderson (1998). Both proposed that model selection, rather than hypothesis testing, is the mechanistic key to the approach originally proposed by Chamberlin and extended by Platt. Chamberlin and Platt initially proposed the philosophy and rationale for using multiple hypotheses. In this chapter we describe some of the tools outlined by Burnham and Anderson (1998) to provide a strength of evidence for multiple models rep-

resenting a set of hypotheses. What we cannot supply here is the intuition, imagination, and knowledge needed to actually develop the candidate hypotheses. These hypotheses can come only from resource managers and researchers creatively merging information about their system with possible explanations of the phenomenon of interest. (See also Chapter 11 in this volume.)

THE MODEL SELECTION APPROACH

The model selection approach in statistics has a number of key components:

- Developing an a priori set of candidate models that represent candidate hypotheses
- Assessing goodness-of-fit of data to the assumptions of the underlying statistical model that forms the "backbone" of the models being considered (such as linear regression)
- Fitting the candidate models to the set of data using least squares or maximum likelihood
- Ranking the candidate models in terms of their estimated Kullback-Leibler information
- Examining the highest-ranking models for inference based on strength of evidence using statistical tools derived from Kullback-Leibler information
- Estimating the amount of variation in the data explained by the models selected for inference
- Making formal inferences from the best model in the set and, in some cases, making multimodel inferences

We will discuss each of these components in more detail with an example. Further details on multimodel inference can be found in Buckland et al. (1997) and Burnham and Anderson (1998).

A Unifying Example

To demonstrate the methodology and utility of the model selection approach, consider a fictitious example based on simulated data. Simulated data, where the generating model and its parameters are known, are useful when we want to examine how well different statistical methods perform. For our fictitious example, we generated 10 years of reproductive data from the brown owl on an isolated island in the South Pacific. (Neither the owl nor the place actually exists.) There are only 250 pairs of owls on this island, and the population size has varied little during the 10-year period. Each

pair of owls occupies a distinct territory, and for each territory in each year the reproductive outcome is either 0, 1, 2, or 3 young fledged. The distribution of the 0s, 1s, 2s, and 3s, however, does not follow any categorical distribution (such as a Poisson distribution). For each year, the mean annual reproductive output (R) is based on the average across the territories where the sampling variances of R behave similar to a Poisson distribution in that they increase as the mean increases. Thus reproductive output within each year is nonnormally distributed, and the sampling variances of R are heterogeneous (not equal across years).

The people of the island where the owl occurs practice slash-and-burn agriculture—that is, they periodically cut small patches of forest and burn the plot to grow crops. A scientist has been working in the field on the brown owl for the past 10 years and has sampled the same 50 territories (out of the total 250 territories) each year. By sampling the same 50 territories each year, the field scientist has introduced a lack of independence in the estimates of R between years: the outcome on a territory in a given year may affect the outcome there in the following year. This lack of independence cannot be resolved in the design of the field study, however, because of logistic constraints. In examining the preliminary data, the field scientist has noticed that reproductive output has declined over time (Figure 5.1) and consults with other scientists about the potential causes for this decline.

One of the consulting scientists, a parasitologist, believes that an increase in a certain bloodsucking fly (the hippoboscid fly) that lives in the plumage of birds is the cause for the decline. He hypothesizes that the agricultural

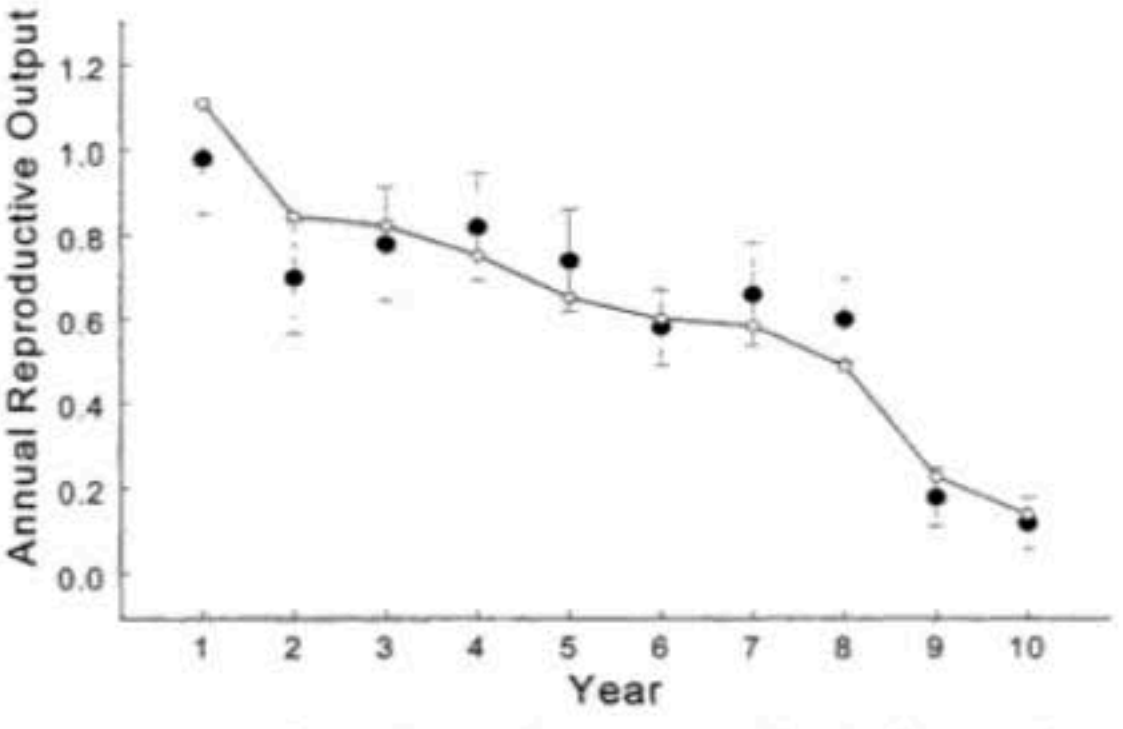


Figure 5.1. Estimates of annual reproductive output for the fictitious brown owl (solid dots) in relation to the known values (open dots connected by solid line) from which the estimates were drawn. Vertical bars are ± one standard error.

practices of the indigenous people have limited the number of cavity nests that the owl uses, thus exposing the young to increased parasite populations as the same nests are used repeatedly. The field scientist had routinely collected parasites while banding young and was able to develop an annual variable called PARA (the mean number of hippoboscid flies per young). A second scientist, a conservation biologist, believes that increasing smoke from the fires used in the slash-and-burn agriculture is negatively affecting survival of the young in the nest prior to fledging. To examine this idea, he developed a variable called FIRE (the acreage burned each year). The field scientist, however, believes that the cause is weather-related; she has noticed that field seasons have been wetter in the last few years she has conducted fieldwork. She hypothesizes that increased precipitation during the nesting season affects the ability of the nocturnal parents to hunt and provide food for the young. A third scientist, an ornithologist, does not think there is a single cause. Based on conversations with the field scientist and the parasitologist, he proposes that it is a combination of precipitation and parasites. Thus there is some disagreement about the cause of the decline in R for the brown owl. Each scientist, therefore, has developed a hypothesis based on the literature or experience working with the brown owl.

We know the answer to this problem because we simulated the data. In the generating model, mean annual reproductive output was perfectly correlated with precipitation during the breeding season and only slightly correlated with fire and parasites. Thus the generating model is one where reproductive output has a perfect negative linear relationship with precipitation during the breeding season. Expressed as a linear regression, the model used to generate our data is

$$E(R) = \beta_0 - \beta_1 \text{ (PRECIP)} = 2.00 - 2.00 \text{ (PRECIP)}$$

(with no residual error), where 2.00 is the intercept (β_0) and 2.00 is also the slope on precipitation (β_1). In the real world, none of the scientists know this. In the following sections we outline the process of how the conflicting hypotheses of the scientists were examined in the case of the brown owl.

The Set of Candidate Models

Often wildlife studies involve the collection of data on numerous variables that are then exhaustively searched for patterns and "significant" relationships to some response variable (reproductive output, in our example). This type of analysis is sometimes called *data dredging* (Anderson and Burnham 1999a) and often results in spurious conclusions and selection of fitted models that have more variables than are supported by the data. A valid alternative is to develop a priori hypotheses before analyzing the data

(Burnham and Anderson 1998; Anderson and Burnham 1999a). These a priori hypotheses are based on biological and management considerations and represent plausible candidate models. In fact, they are the multiple working hypotheses that Chamberlin (1890) advocated over a century ago.

In our brown owl example, the ideas advocated by the different scientists were initially expressed as concepts that we then translated into statistical models in the form of linear regression models (Table 5. 1). Apart from their own ideas, the four scientists also agreed to include two additional models of reproductive output. Trend Model 1 represents the hypothesis that reproductive output is declining but not for the reasons suggested by

Table 5.1. Candidate Set of Models Developed to Explain Apparent Decline in Brown Owl Reproduction

Advocate	*Concept*	*Statistical model*
Field scientist	Increased precipitation during the nesting season (PRECIP) is negatively associated with annual reproductive output	$R = \beta_0 - \beta_1$ (PRECIP)
Parasitologist	Increased parasite loads on young (PARA) are negatively associated with annual reproductive output	$R = \beta_0 - \beta_1$ (PARA)
Conservation biologist	Increased acreage burned for farming (FIRE) is negatively associated with annual reproductive output	$R = \beta_0 - \beta_1$ (FIRE)
Ornithologist	The combination of increased precipitation and increased parasite loads is negatively associated with annual reproductive output	$R = \beta_0 - \beta_1$ (PRECIP) $- \beta_2$ (PARA)
Trend Model 1	There is a negative linear trend (T) in reproductive output over time but not due to the variables proposed	$R = \beta_0 - \beta_1$ (T)
Trend Model 2	There is no trend over time in reproductive output	$R = \beta_0$

Note: The models are presented as the concepts of four scientists and translated into a priori statistical models concerning the causes for decline in annual reproductive output (R) in the fictitious brown owl over a 10-year period.

the four scientists. Trend Model 2 expresses the hypothesis that reproductive output is approximately constant over time. Considered together, the six models presented in Table 5.1 are based on biological and management considerations prior to conducting the analysis and represent different hypotheses about the decline in brown owls during the 10-year study period.

What would have happened if the scientists had decided to employ a data-dredging strategy rather than selecting a set of carefully considered, a priori candidate models? To explore this possibility, we subjected the simulated data to forward, backward, and stepwise regression procedures where each variable was examined for "significance" based on an arbitrary α level (in this case, $\alpha = 0.10$, the default set by the computer program). Although widely used, these procedures lack any theoretical justification (referred to as ad hoc). The resulting model from all three procedures included both precipitation during the breeding season (PRECIP) and parasite loads (PARA):

$$R = \beta_0 - \beta_1 (\text{PRECIP}) - \beta_2 (\text{PARA})$$

Although it includes one of the variables from the generating model, it contains more parameters than can be accounted for by the data. Thus the model selected by data dredging would lead us to believe that the hypothesis espoused by the ornithologist (see Table 5.1) is the most plausible when in fact it is not the model that was used to generate the sample data.

Goodness-of-Fit

All statistical models used to estimate parameters rely on some underlying distribution. (See Chapter 3 of this volume.) The typical use of linear regression, for example, whether based on likelihood or least-squares estimators, assumes that the residuals are independent, normally distributed, and have constant variance (Graybill and Iyer 1994). Goodness-of-fit testing examines whether the data being considered fit these underlying assumptions. Under the model selection approach, goodness-of-fit is examined with the model having the largest number of parameters. When assumptions underlying the statistical model are known to be violated, these violations must be accounted for. In the brown owl example, we simulated the data to violate two assumptions underlying the typical use of linear regression: independence of observations and homogeneity of variances. Lack of independence will not affect parameter estimates, but estimated standard errors will appear more precise than they should be if the lack of independence between years is not accounted for. One way we can account for these two violations is by using likelihood-based, mixed-effects models

(Littell et al. 1996)—models that include both fixed effects (precipitation, parasite loads, and so forth) and random effects (year, territory). With this type of analysis, the covariances between years (to account for lack of independence) and the proportionality of means with the sampling variances (to account for heterogeneity of variances) can be estimated and incorporated into the analysis. Thus what would be assumption violations in a simple analysis are eliminated by using an appropriate generalized analysis. The lack of normality cannot be dealt with in this manner. Instead we must rely on the robustness of regression and analysis of variance even to severe departures from normality, as is the case with the simulated count data for the brown owl (White and Bennetts 1996).

Model Selection Criteria

Once a set of candidate models is proposed, how do we select the one that best explains our data? We have seen that the ad hoc procedures do not work well. Ideally we want some measure that allows us to state that a certain model is preferred, and by how much, over other models to explain a set of data. There are two further desirable properties of such a model selection criterion: it should be objective (the investigator cannot change the outcome by adjusting subjective inputs to the analysis), and it should have a sound theoretical basis. Objectivity is a key component of this criterion because of the tenets of science. The stepwise model selection procedure we used previously with our simulated data, for example, would not qualify as an objective model selection procedure; a different model would be selected if we changed the α level from 0.10 to 0.20 or to 0.05. Anderson and Burnham (1999b) and Burnham and Anderson (1998) have reviewed a number of alternatives for model selection and conclude that Akaike's Information Criterion (AIC) (Akaike 1973) is an appropriate criterion. AIC is mathematically defined as

$$\text{AIC} = -2 \log_e \mathcal{L}(\text{parameters} \mid \text{data}) + 2(\text{number of parameters})$$

The first component of AIC represents how well the model fits the data. Recall the definition of the likelihood presented earlier in this chapter: $\mathcal{L}(\underline{\theta} \mid X, \text{model})$, the likelihood of the parameters given the data X and the model (hypothesis). Here the hypothesis is represented by the parameters in the statistical model that was translated from the conceptual ideas of the scientists. The likelihood expressed for the model suggested by the field scientist, for example, would be written simplistically as

$$\mathcal{L}\{\beta_0, \beta_1, \sigma^2 \mid \text{data and model } R = \beta_0 - \beta_1 (\text{PRECIP})\} \qquad [5.1]$$

which can be translated as the "likelihood of parameters β_0, β_1, and σ^2 given

the data and model $R = \beta_0 - \beta_1$ (PRECIP)." The higher the value of the likelihood, the better the model's fit to the data. The second component of AIC represents a penalty in terms of the number of parameters in the model. Why penalize based on the number of parameters? The reason is based on the principle of parsimony (see Burnham and Anderson 1992), which represents a tradeoff between models having too few parameters and models that have too many. A model such as $\{R = \beta_0\}$ may have too few parameters (only two: β_0 and σ^2), and the estimates of R would be biased relative to the true values. The estimates would be precise, however, because all of the data are being used to estimate the two parameters. A model that estimates parameters for each of the 10 years, such as $R = \beta_0 + \beta_1$(Year 2) $+ \cdots + \beta_9$ (year 9) may be less biased with an estimate for each year, but it would also be much less precise because the data are divided to estimate 11 parameters (ten β plus σ^2). Thus as the number of parameters increases, bias is reduced but precision is lost. As the number of parameters decreases, bias worsens as precision improves. AIC balances this tradeoff between bias and precision. Within the set of candidate models, the model having the lowest AIC is the "best" model in terms of estimated Kullback-Leibler information. That is, AIC is an estimate of the expected, relative Kullback-Leibler information lost when a model is used to approximate reality. These ideas of parsimony and penalty for too many parameters in a model become by-products of estimating Kullback-Liebler information using AIC.

When $n/K < 40$ (where K is the number of parameters and n is the sample size), AIC poorly estimates Kullback-Leibler. In this case, a second-order version of AIC, AIC_c, should be used (Hurvich and Tsai 1989):

$$AIC_c = AIC + \frac{2K(K+1)}{n-K-1} \qquad [5.2]$$

In our example, we would use AIC_c because the sample size we are ultimately dealing with is $n = 10$ years.

Ranking of Candidate Models

Both AIC and AIC_c can be used to rank models within a set of candidate models such as these proposed in Table 5.1. Once ranked, however, how good is the "best" model relative to the second-ranked model or even the third-ranked model? To resolve this question, there is an additional tool, Akaike weights (Buckland et al. 1997), that can be assigned to each model i in a set containing M candidate models:

$$w_i = \frac{\exp(-\frac{\Delta_i}{2})}{\sum_{m=1}^{M} \exp(-\frac{\Delta_i}{2})} \qquad [5.3]$$

where Δ_i is the difference between model i and the model with the lowest AIC (or AIC_c) value. The weights for all models in the candidate set sum to 1 and, for each model i, represent the likelihood of that model given the data. This idea of the likelihood of the model given the data, and hence these model weights, has been suggested for many years by Akaike (1978, 1979, 1980, 1981, 1983). Strength of evidence for model i versus model j can be expressed as the ratio w_i/w_j. This ratio addresses the question "How much more likely is model i than model j given these data?"

Now we have the tools to rank and examine the likelihood of each model in our candidate set for the brown owl. The results of this exercise are shown in Table 5.2. First we were able to rank the proposed hypotheses in terms of their plausibility, using AIC_c, and find that the model stemming from the hypothesis proposed by the field scientist $\{R = \beta_0 - \beta_1 (\text{PRECIP})\}$ is the most plausible given the data. Although this model also happens to be the generating model ("truth"), we reiterate that the objective when using AIC is to select the best approximating fitted model given the data. What was the likelihood of this model relative to the other models in the candidate set—or, in other words, were any of the other models similarly likely to be the best model selected by AIC_c? To answer this question we examine the Akaike weights in Table 5.2 and find that the best model is highly likely as well, with no other models coming close in terms of their relative likelihood. Remember that the Akaike weights for all the models in the candidate set sum to 1. Therefore, the best model has almost all (0.933) of the weight associated with it. In terms of strength of evidence, the best model is 23 times (0.933/0.040) more likely than the second-ranked model. Thus we conclude that the model representing the hypothesis proposed by the field scientist is the most plausible. But how good a model is it? It may be the best model out of the set of models proposed by the scientists. But if it does not explain much of the variation in annual reproductive output, it is not a very good model. We address this question in the next section.

Table 5.2. Ranking and Weighting of the Set of Candidate Models

Model	$K^{\dagger}$	AIC_c	*Rank*	Δ_i	*Akaike weights*
$R = \beta_0 - \beta_1$ (PRECIP)	3	1053.80	1	0.00	0.933
$R = \beta_0 - \beta_1$ (PRECIP) $- \beta_2$ (PARA)	4	1060.10	2	6.30	0.040
$R = \beta_0 - \beta_1$ (T)	3	1062.10	3	8.30	0.015
$R = \beta_0$	2	1062.60	4	8.80	0.011
$R = \beta_0 - \beta_1$ (FIRE)	3	1068.70	5	14.90	0.001
$R = \beta_0 - \beta_1$ (PARA)	3	1069.80	6	16.00	0.000

† K is the number of parameters—in this case, the number of β plus σ^2.

where Δ_i is the difference between model i and the model with the lowest AIC (or AIC_c) value. The weights for all models in the candidate set sum to 1 and, for each model i, represent the likelihood of that model given the data. This idea of the likelihood of the model given the data, and hence these model weights, has been suggested for many years by Akaike (1978, 1979, 1980, 1981, 1983). Strength of evidence for model i versus model j can be expressed as the ratio w_i/w_j. This ratio addresses the question "How much more likely is model i than model j given these data?"

Now we have the tools to rank and examine the likelihood of each model in our candidate set for the brown owl. The results of this exercise are shown in Table 5.2. First we were able to rank the proposed hypotheses in terms of their plausibility, using AIC_c, and find that the model stemming from the hypothesis proposed by the field scientist $\{R = \beta_0 - \beta_1 (\text{PRECIP})\}$ is the most plausible given the data. Although this model also happens to be the generating model ("truth"), we reiterate that the objective when using AIC is to select the best approximating fitted model given the data. What was the likelihood of this model relative to the other models in the candidate set—or, in other words, were any of the other models similarly likely to be the best model selected by AIC_c? To answer this question we examine the Akaike weights in Table 5.2 and find that the best model is highly likely as well, with no other models coming close in terms of their relative likelihood. Remember that the Akaike weights for all the models in the candidate set sum to 1. Therefore, the best model has almost all (0.933) of the weight associated with it. In terms of strength of evidence, the best model is 23 times (0.933/0.040) more likely than the second-ranked model. Thus we conclude that the model representing the hypothesis proposed by the field scientist is the most plausible. But how good a model is it? It may be the best model out of the set of models proposed by the scientists. But if it does not explain much of the variation in annual reproductive output, it is not a very good model. We address this question in the next section.

Table 5.2. Ranking and Weighting of the Set of Candidate Models

Model	$K^{\dagger}$	AIC_c	*Rank*	Δ_i	*Akaike weights*
$R = \beta_0 - \beta_1$ (PRECIP)	3	1053.80	1	0.00	0.933
$R = \beta_0 - \beta_1$ (PRECIP) $- \beta_2$ (PARA)	4	1060.10	2	6.30	0.040
$R = \beta_0 - \beta_1$ (T)	3	1062.10	3	8.30	0.015
$R = \beta_0$	2	1062.60	4	8.80	0.011
$R = \beta_0 - \beta_1$ (FIRE)	3	1068.70	5	14.90	0.001
$R = \beta_0 - \beta_1$ (PARA)	3	1069.80	6	16.00	0.000

† K is the number of parameters—in this case, the number of β plus σ^2.

How Good Is the Best Model?

There are a number of ways to assess how good a model is in terms of explaining the data. In the model selection approach (as in hypothesis testing), it is plausible to have inappropriate models in the candidate set; AIC will select the best of these models. Although the best model selected by AIC will be the best of the lot examined, it may still be a poor model in terms of explaining the data. Therefore, we want to further examine our best selected model for the brown owl. First we want to examine the precision of the estimated regression coefficient for PRECIP ($\hat{\beta}_1$) and see whether the 95 percent confidence interval for this parameter overlaps zero. If the interval overlaps zero considerably, there is little support for the trend in reproductive output being due to changes in precipitation. In the case of our best model $\{R = \beta_0 - \beta_1 \text{ (PRECIP)}\}$, $\hat{\beta}_1 = -0.196$ with $\text{SE}(\hat{\beta}_1) = 0.021$ and 95 percent confidence intervals of −0.237, −0.155. In this case, the parameter both is precise and has a 95 percent confidence interval that indicates the slope is quite different from zero.

Next we want to find out how much variation in reproductive output is explained by the best model. To do this we use a variance-components analysis (see Chapter 3 in this volume). Here we are interested in the total temporal variation ($\sigma^2_{\text{temporal}}$) and how much of that variation is explained by the best model (σ^2_{model}). The percentage of variation explained by the best model can then be calculated as

$$\% \text{ variation explained} = \frac{\hat{\sigma}^2_{\text{model}}}{\hat{\sigma}^2_{\text{temporal}}} \times 100 \qquad [5.4]$$

Without going into details about the variance-components analysis, we simply note that it can easily be done as a by-product of our mixed-effects modeling with the simulated brown owl data (see Franklin et al. 2000). The best model selected by AIC_c explains all of the temporal variation (Table 5.3) after separating out the annual sampling variance. This model also has

Table 5.3. Variance-Components Analysis for Total Temporal Variation and the Amount and Percentage of Variation Explained by Various Models

Type of variation	$\hat{\sigma}^2$	*%*†
Total temporal ($\sigma^2_{\text{temporal}}$)	0.0619	–
Explained by model $\{\beta_0 - \beta_1 \text{ (PRECIP)}\}$	0.0619	100.0
Explained by model $\{\beta_0 - \beta_1 \text{ (PRECIP)} - \beta_2 \text{ (PARA)}\}$	0.0619	100.0

†Percentage of temporal variation explained by the model.

precise estimates different from zero and explains a considerable amount of the process variation in annual reproductive output for the brown owl. Thus it is a good model.

Moreover, the model selected by using stepwise procedures also explains a large amount of variation (Table 5.3). This illustrates why using the amount of variance explained is a poor criterion for selecting the best model. As parameters are added to a model—even totally useless parameters—the amount of variance explained will increase. We know that the variable PARA is essentially random noise in our simulated data and adds little to explaining the data because we made it that way. But if we had chosen it for inference, we would be ascribing some biological meaning to the presence of parasites and their effect on brown owls.

Advantages of the Approach

The increased use of statistical model selection to explain data is relatively new to natural resource management. In this chapter we have outlined the concepts and procedures used in the model selection approach based on Kullback-Leibler information and likelihood-based inference to select the best model from a set of candidates. We also contrasted this model selection approach with traditional null hypothesis testing. For further clarification, we demonstrated the model selection approach through an example of simulated data where we knew the generating model. For the philosophy, statistical details, and theory behind the model selection approach we have outlined here, we urge readers to review Burnham and Anderson (1998).

The statistical model selection approach not only eliminates the arbitrary limitations of null hypothesis testing, such as significance level and tests of null hypotheses, but also embraces the concept of multiple working hypotheses as first proposed by Chamberlin (1890). Rarely is there a single, simple, true model that explains a natural phenomenon. To avoid being misled by individual biases, scientists and managers should develop a set of candidate models to explain the phenomenon of interest. Developing a set of plausible, creative candidate models is the key to the quality of inference that can be drawn from the data. Incorporating ideas from the resource managers and researchers involved with the problem is critical to the development of the best set of models.

Statistical model selection offers a valid, comprehensive approach to evaluating the evidence for alternative scientific hypotheses given the data. Whatever information is supported by the data, the entire set of candidate models is incorporated in the final inference—through selection of the best model, ranking of candidate models, and strength of evidence for the best and competing models in the candidate set. Such an approach yields valid

inferences following objective analysis of a broad spectrum of a priori hypotheses.

ACKNOWLEDGMENTS

We thank Mark Bakeman, Monica Bond, and Tom Ryan for their excellent comments on previous drafts of this manuscript.

REFERENCES

Akaike, H. 1973. Information theory as an extension of the maximum likelihood principle. Pages 267–281 in B. N. Petrov and F. Csaki, eds., *Second International Symposium on Information Theory.* Budapest: Akademiai Kiado.

———. 1978. A Bayesian analysis of the minimum AIC procedure. *Annals of the Institute of Statistical Mathematics* 30:9–14.

———. 1979. A Bayesian extension of the minimum AIC procedure of autoregressive model fitting. *Biometrika* 66:237–242.

———. 1980. Likelihood and the Bayes procedure. Pages 143–203 in J. M. Bernardo, M. H. De Groot, D.V. Lindley, and A. F. M. Smith, eds., *Bayesian Statistics.* Valencia: University Press.

———. 1981. Modern development of statistical methods. Pages 169–184 in P. Eykhoff, ed., *Trends and Progress in System Identification.* Paris: Pergamon Press.

———. 1983. Information measures and model selection. *International Statistical Institute* 44:277–291.

Anderson, D. R., and K. P. Burnham. 1999a. General strategies for the analysis of ringing data. *Bird Study* 46(suppl.):S261–S270.

———. 1999b. Understanding information criteria for selection among capture–recapture or ring recovery models. *Bird Study* 46 (suppl.):S14–S21.

Buckland, S.T., K. P. Burnham, and N. H. Augustin. 1997. Model selection: An integral part of inference. *Biometrics* 53:603–618.

Burnham, K. P., and D. R. Anderson. 1992. Data-based selection of an appropriate biological model: The key to modern data analysis. Pages 16–30 in D. R. McCullough and R. H. Barrett, eds., *Wildlife 2001: Populations* London: Elsevier Applied Science.

———. 1998. *Model Selection and Inference: A Practical Information-Theoretic Approach.* New York: Springer-Verlag.

Chamberlin, T. C. 1890. The method of multiple working hypotheses. *Science* (old series) 15:92. Reprinted in 1965 in *Science* 148:754–759.

Edwards, A. W. F. 1992. *Likelihood.* Expanded ed. Baltimore: Johns Hopkins University Press.

Fisher, R. A. 1925. Theory of statistical estimation. *Proceedings of the Cambridge Philosophical Society* 22:700–725.

———. 1958. *Statistical Methods for Research Workers.* 13th ed. New York: Hafner.

Franklin, A. B., D. R. Anderson, R. J. Gutiérrez, and K. P. Burnham. 2000. Climate,

habitat quality, and fitness in northern spotted owl populations in northwest California. *Ecological Monographs* 70: 539–590.

Graybill, F. A., and H. K. Iyer. 1994. *Regression Analysis: Concepts and Applications.* Belmont, Calif.: Duxbury.

Hilborn, R., and M. Mangel. 1997. *The Ecological Detective: Confronting Models with Data.* Monographs in Population Biology 28. Princeton: Princeton University Press.

Hurvich, C. M., and C.-L. Tsai. 1989. The impact of model selection on inference in linear regression. *American Statistician* 44:214–217.

Kullback, S., and R. A. Leibler. 1951. On information and sufficiency. *Annals of Mathematical Statistics* 22:79–86.

Linkhart, H., and W. Zucchini. 1986. *Model Selection.* New York: Wiley.

Littell, R. C., G. A. Milliken, W. W. Stroup, and R. D. Wolfinger. 1996. *SAS System for Mixed Models.* Cary, N.C.: SAS Institute.

Malakoff, D. 1999. Bayes offers a "new" way to make sense of numbers. *Science* 286: 1460–1464.

Maurer, B. A. 1998. Ecological science and statistical paradigms: At the threshold. *Science* 279:502–503.

McCullagh, P., and J. A. Nelder. 1989. *Generalized Linear Models.* 2nd ed. London: Chapman & Hall.

Miller, A. J. 1990. *Subset Selection in Regression.* London: Chapman & Hall.

Neyman, J., and E. S. Pearson. 1933. On the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society Series A* 231:289–337.

Ott, R. L. 1993. *An Introduction to Statistical Methods and Data Analysis.* Belmont, Calif.: Duxbury.

Platt, J. R. 1964. Strong inference. *Science* 146:1–7.

Romesburg, H. C. 1991. On improving the natural resources and environmental sciences. *Journal of Wildlife Management* 55:744–756.

Royall, R. 1997. *Statistical Evidence: A Likelihood Paradigm.* London: Chapman & Hall.

Schweder, T. 1988. A significance version of the basic Neyman-Pearson theory for scientific hypothesis testing. *Scandinavian Journal of Statistics* 15:225–242.

White, G. C., and R. E. Bennetts. 1996. Analysis of frequency count data using the negative binomial distribution. *Ecology* 77:2549–2557.

Zimmerman, M. 1995. *Science, Nonscience, and Nonsense: Approaching Environmental Literacy.* Baltimore: Johns Hopkins University Press.